📌 Executive Summary – Telecom Customer Churn Analysis

This analysis explores customer behavior and churn patterns within a telecom dataset consisting of **7,043 customers**. The goal is to understand **which customer groups are most likely to leave the company**, identify trends in service usage, and highlight actionable insights that can guide retention strategies.

The study uses visual exploration through count plots, distribution plots, and categorical summaries.

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1. Customer Profile Overview

Gender Distribution

- Male and female customers are almost **equally represented (≈50% each)**.
- **Churn rates are similar**, indicating that gender does **not significantly influence churn**.

Senior Citizens

- Only **16%** of the customers are senior citizens (65+).
- However, the churn rate among seniors is **much higher (≈41%)** compared to non-seniors (≈26%).
- **Insight:** Senior citizens are a high-risk segment needing targeted retention efforts.

Dependents & Partners

- **30%** of customers have dependents, and they tend to churn **less frequently**.
- **52%** have partners, and this group also shows a **lower churn rate**.
- **Insight:** Customers with family responsibilities are more stable.

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2. Tenure Patterns (Customer Loyalty)

- Tenure ranges from 1 to 72 months.
- The highest churn occurs in the **first 6 months**, where churn exceeds **45%**.
- Customers with **long-term tenure (over 2 years)** have a churn rate below **10%**.
- **Insight:** The first six months are critical in customer retention. Improving onboarding and early engagement can reduce churn significantly.

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3. Service Usage Patterns

Phone Services

- Most customers (~90%) have a phone service.
- Churn is higher among customers with **only phone service and no internet**.

Multiple Lines

- Around **45%** have multiple phone lines.
- Slightly higher churn among multi-line customers, possibly due to **bundle dissatisfaction**.

Internet Service Type

- **Fiber optic users form ~44%** of the base but account for **61% of total churn**.
- DSL users churn much less.
- **Insight:** Fiber optic users are the highest-churn segment — likely due to dissatisfaction with speed, pricing, or reliability.

Value-Added Services (OnlineSecurity, Backup, TechSupport, etc.)

Across all optional services:

- Customers **without protection/security add-ons churn at 40–60% rates**.
- Customers **with these services churn <20%**.
- **Insight:** Security, backup, and tech-support services increase stickiness. Upsell strategies can directly reduce churn.

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4. Contract, Billing, and Payment Insights

Contract Type

- Month-to-month contracts make up **56%** of customers.
- **Churn rate: ~43%**, the highest among all groups.
- One-year and two-year contracts have **much lower churn (<12%)**.
- **Insight:** Longer contracts significantly improve customer retention.

Paperless Billing

- **57%** of customers use paperless billing.

- Their churn rate is **higher (~33%)**, possibly due to digital-only promotions or ease of switching.

Payment Method

- Electronic check users have the **highest churn (≈45%)**.
- Customers using credit card, bank transfer, or mailed checks churn much less.
- **Insight:** Payment mode is a strong predictor of churn.

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5. Financial Insights

Monthly Charges

- Higher monthly charges correlate with **higher churn**.
- Customers paying **above $80/month** show churn rates exceeding **45%**.

Total Charges

- Low total charges (new customers) = **high churn**
- High total charges (long relationship) = **low churn**
- **Insight:** New customers are more likely to churn; long-term customers are loyal.

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6. Overall Churn Summary

- **Total churn rate: ~26–27%**.
- Key churn drivers:
    1. **Month-to-month contracts**
    2. **Fiber optic internet**
    3. **Electronic check payments**
    4. **High monthly charges**
    5. **Lack of security/tech-support add-ons**
    6. **Low tenure (especially under 6 months)**

📌 **Final Takeaways (Most Important Insights)**

✓ Customers most likely to churn:

- New customers
- Fiber optic users
- Month-to-month subscribers
- Electronic check payers
- Users without security or tech-support add-ons

✔ **Customers least likely to churn:**

- Long-term customers (>2 years)
- Two-year contract holders
- Customers with multiple add-on services
- Customers with dependents or partners

✔ **Recommended Actions:**

1. **Introduce retention schemes** for month-to-month customers.
2. **Improve fiber optic customer experience** (speed, support, pricing).
3. **Encourage switching from electronic check to autopay**.
4. **Bundle value-added services** like OnlineSecurity and TechSupport.
5. **Strengthen the first 6-month onboarding journey**